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Clinical: Obstructive Lung Diseases & Restrictive Lung Diseases

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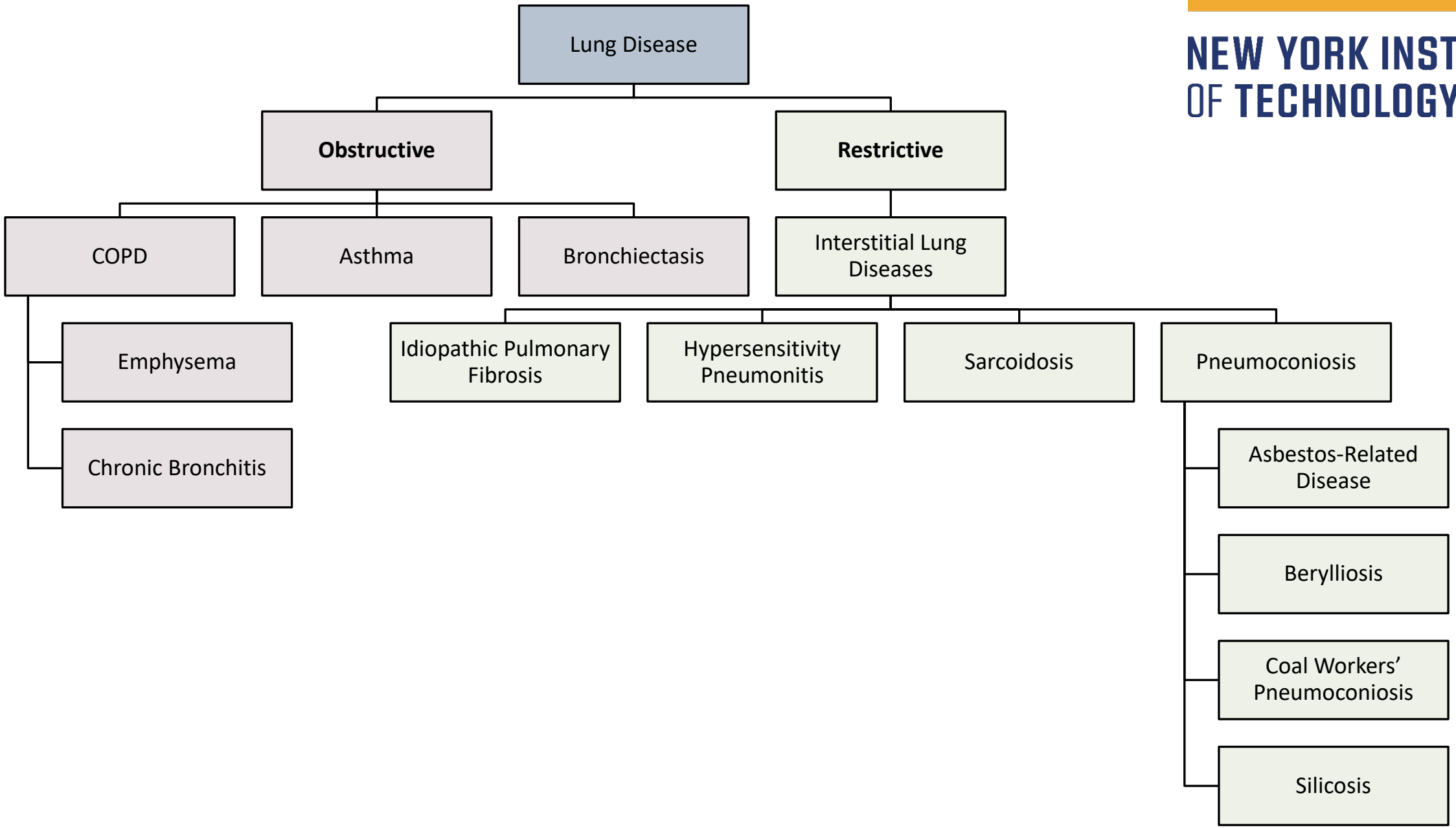
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Learning Objectives

1. Diagnose obstructive versus restrictive lung diseases based on spirometry and flow-volume loops
2. List symptomology and clinical presentations of obstructive and restrictive lung diseases
3. Describe the etiology and pathophysiology of obstructive and restrictive lung diseases
4. Identify diagnostic methodologies and findings of obstructive and restrictive lung diseases
5. Recognize risk factors for obstructive and restrictive lung diseases
6. Select appropriate treatment modalities including pharmacological and non-pharmacological management
7. Predict complications of obstructive and restrictive lung diseases based on evaluation of patient case scenarios

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Patient presents with long term shortness of breath and undergoes spirometry testing, which reveals increased total lung capacity and a decreased forced expiratory volume in one second (FEV1) and forced vital capacity (FVC) with FEV1 is decreased more than FVC.

Is this patient presenting with obstructive or restrictive lung disease pattern?

What is on your differential for this classification of lung disease?

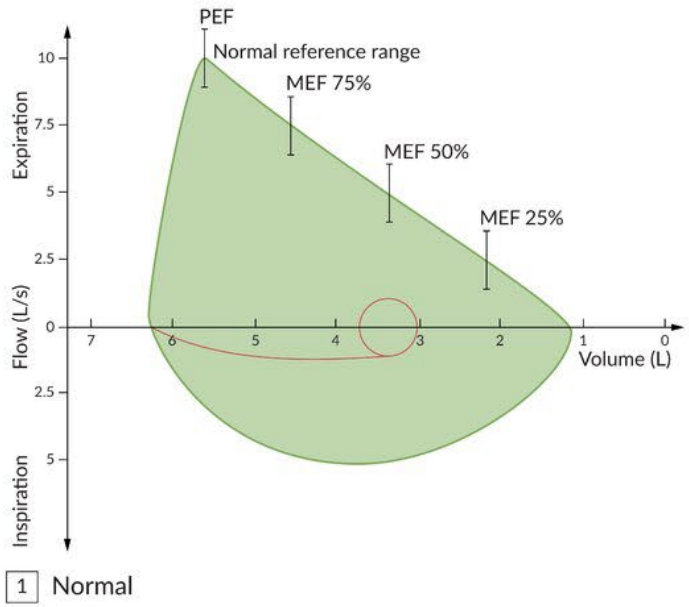
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Is this patient presenting with obstructive or restrictive lung disease pattern?
Obstructive lung disease

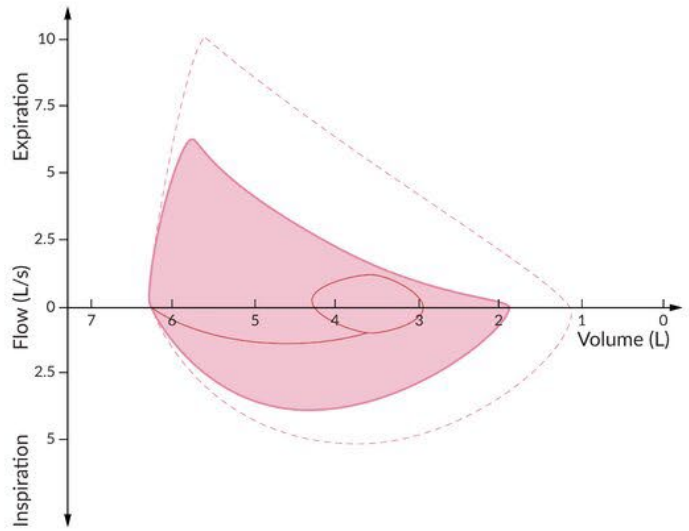
What is on your differential for this classification of lung disease?
COPD including emphysema and chronic bronchitis, asthma, bronchiectasis

Flow-Volume Loops

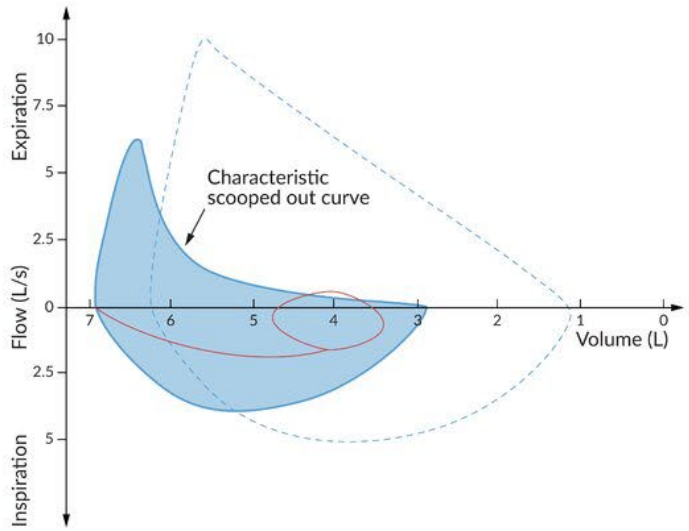
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1 Normal



2 Obstruction in acute asthma



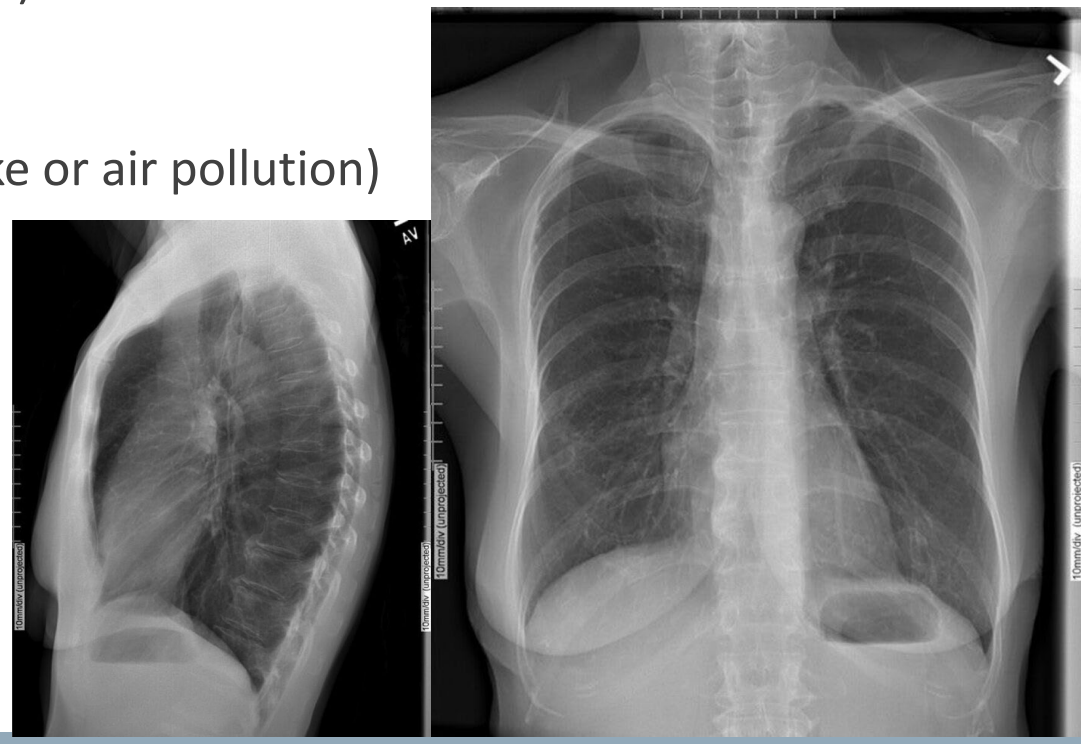
3 Obstruction in emphysema

Flow-Volume Parameter	Obstructive Lung Disease	Restrictive Lung Disease
RV	↑	↓
FRC	↑	↓
TLC	↑	↓
FEV1	↓↓	↓
FVC	↓	↓
FEV1/FVC	↓ [FEV1 decreased more than FVC]	Normal or ↑ [FEV1 decreased proportionately to FVC]

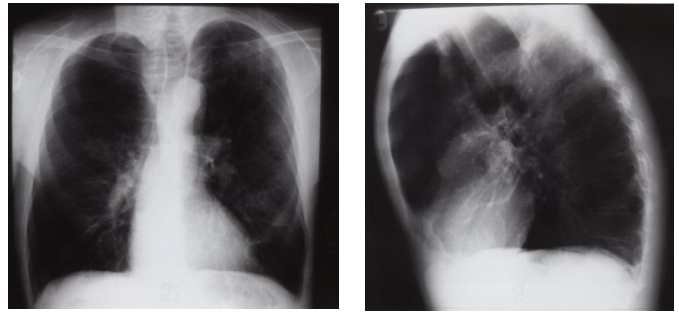
Chronic Obstructive Pulmonary Disease (COPD): Emphysema & Chronic Bronchitis

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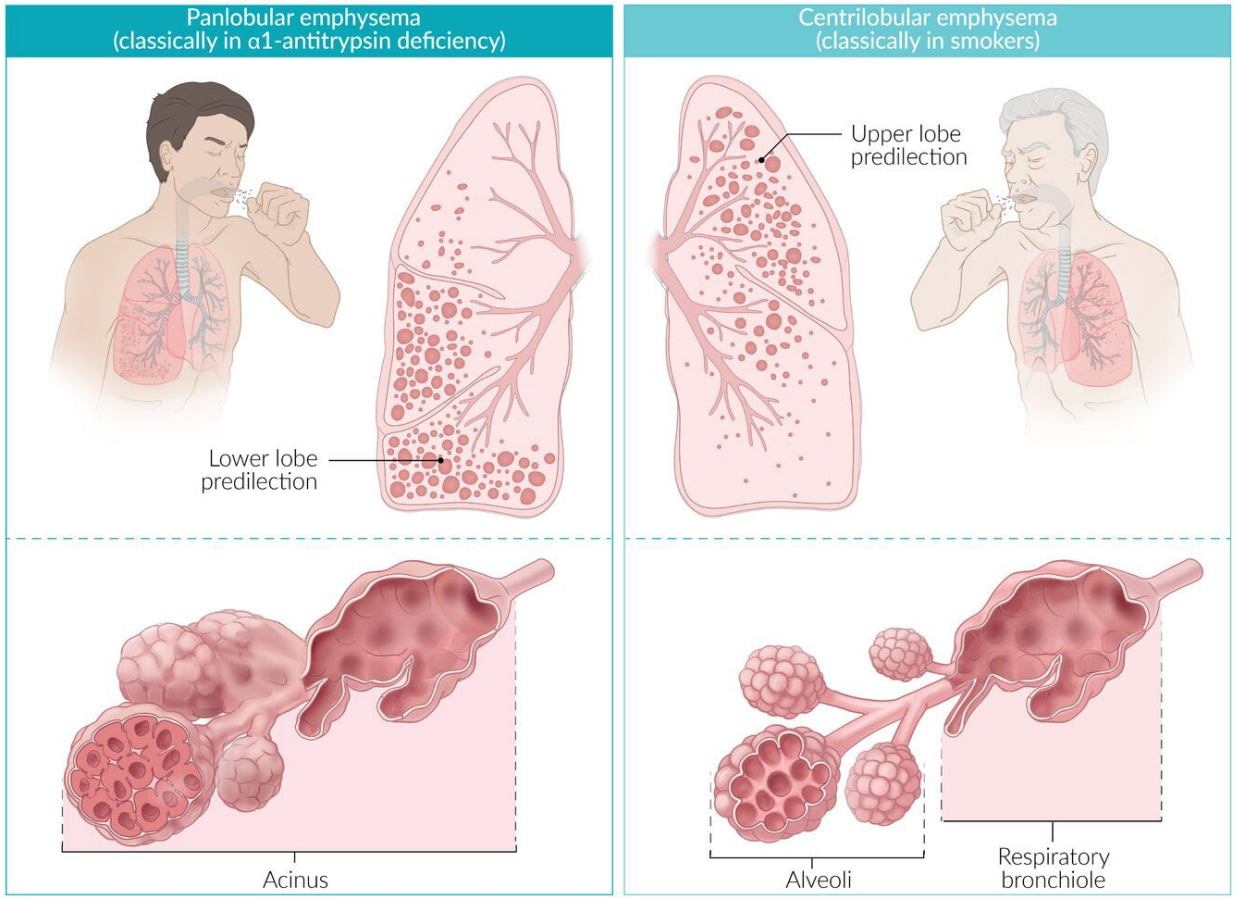
- Persistent respiratory symptoms (cough, dyspnea) resulting from airflow obstruction and alveolar gas exchange abnormalities
- Airflow limitation (postbronchodilator **FEV1:FVC ratio < 0.70**)
- Airway obstruction and parenchymal destruction
- Predominantly caused by inhaled toxins (e.g., tobacco smoke or air pollution)
- **CXR: pulmonary hyperinflation**
  - Increased AP diameter (barrel chest)
  - Flattened diaphragm
  - Widened intercostal spaces
  - Hyperlucency of lung tissue
  - Long and narrow heart shadow



Emphysema “Pink Puffer”

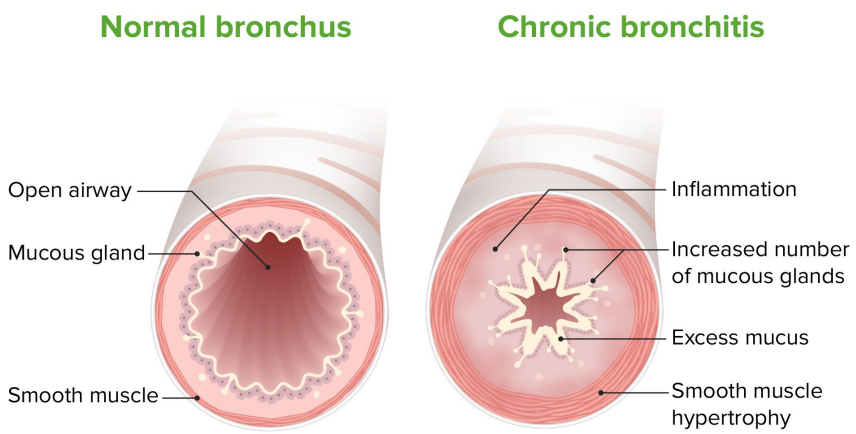


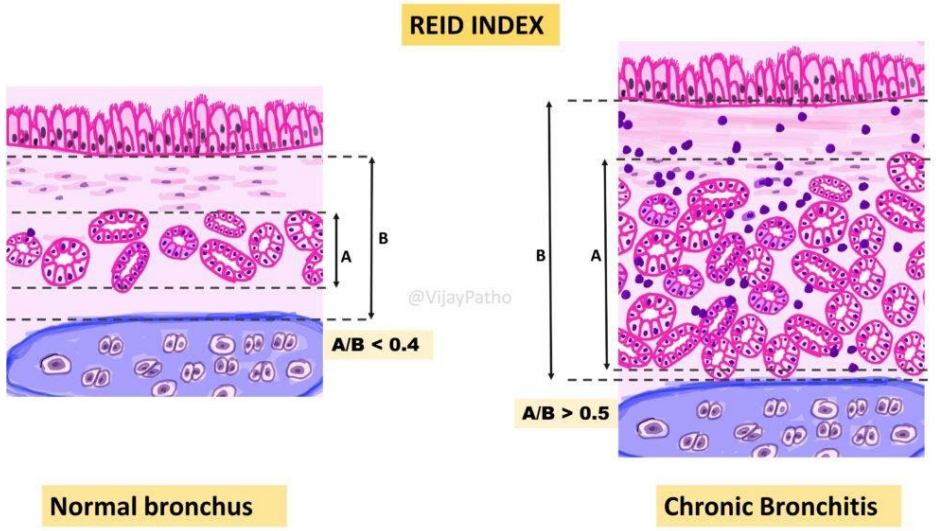
- Presents with **barrel-shaped chest** with prolonged expiration through pursed lips
- Pathophysiology:
  - Permanent **dilatation** of pulmonary air spaces distal to the terminal bronchioles that is caused by the destruction of the alveolar walls and pulmonary capillaries required for gas exchange
  - **Imbalance of proteases and antiproteases** → ↑ elastase activity → ↑ loss of elastic fibers → ↑ lung compliance, ↓ recoil, ↓ DLCO
- Panacinar
  - Associated with **alpha1-antitrypsin deficiency**
  - **Lower lobes**
- Centriacinar
  - Associated with **tobacco smoking**
  - **Upper Lobes**
- CXR: ↑ AP diameter, flattened diaphragm, hyperlucent lungs



Chronic Bronchitis "Blue Bloater"

- Presents with cough, wheezing, crackles, cyanosis, dyspnea, CO_2 retention, secondary polycythemia
- **Diagnostic criteria: productive cough for at least 3 months per year for 2 consecutive years that cannot be explained by an alternative diagnosis**
- Pathophysiology: Hypertrophy and hyperplasia of mucus-secreting glands within the bronchi → **Reid Index >50%** (Normal =<40%)





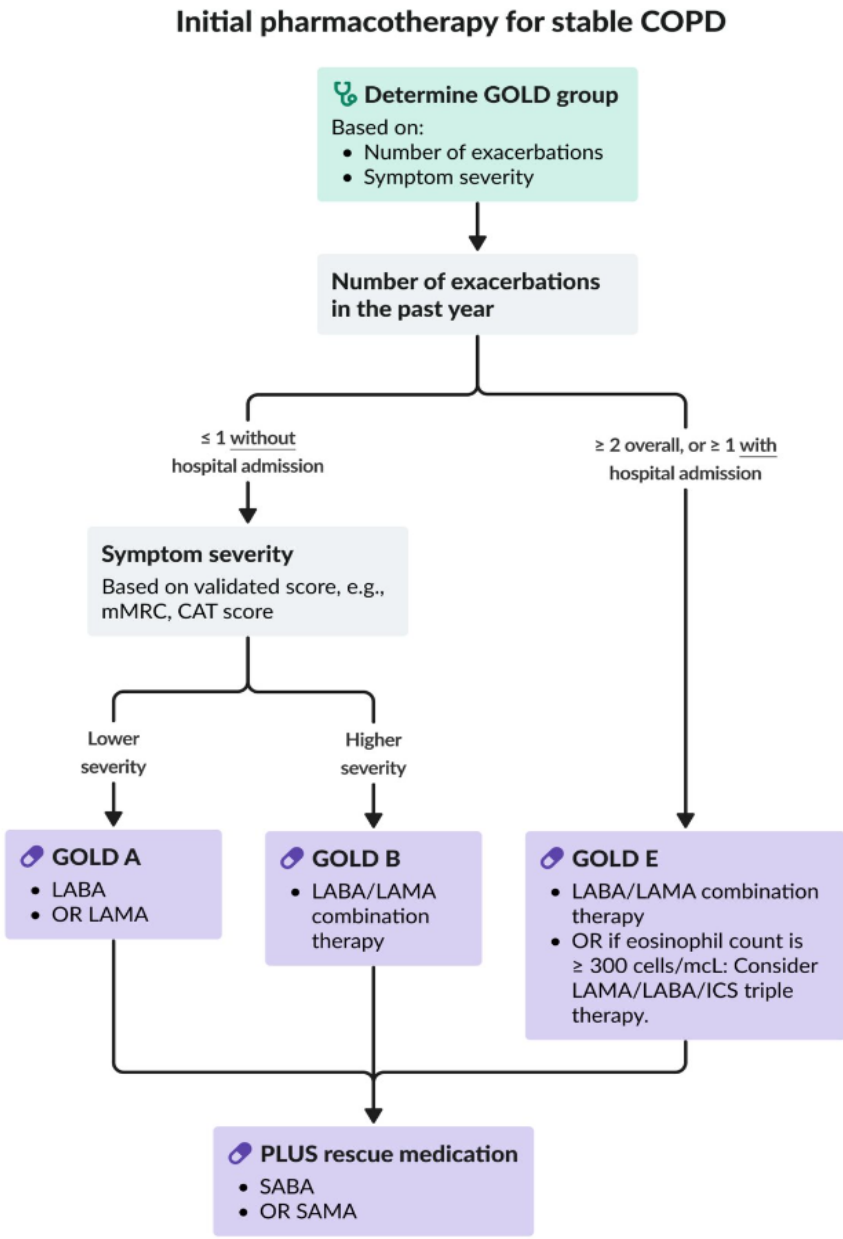
Classification of COPD as Defined by Global Initiative for Chronic Obstructive Lung Disease (GOLD)

	COPD Classification	Definition
Classification of airflow limitation (post-BD FEV_1)	Mild GOLD Stage I	FEV_1 ≥80% predicted
	Moderate GOLD Stage II	FEV_1 ≥50% predicted but <80% predicted
	Severe GOLD Stage III	FEV_1 ≥30% predicted but <50% predicted
	Very severe GOLD Stage IV	FEV_1 <30% predicted
Classification of symptoms/risk of exacerbation mMRC = modified Medical Research Counsel questionnaire	GOLD category A	mMRC 0-1 or CAT <10 (low symptom burden) History of 0 or 1 moderate or severe exacerbations (not leading to hospital admission)
	GOLD category B	mMRC ≥ 2 or CAT ≥10 (higher symptom burden) History of 0 or 1 moderate or severe exacerbations (not leading to hospital admission)
	GOLD category C	mMRC 0-1 or CAT <10 (low symptom burden) History of ≥2 moderate/severe exacerbations or ≥1 exacerbation (leading to hospital admission)
	GOLD category D	mMRC ≥ 2 or CAT ≥10 (higher symptom burden) History of ≥2 moderate/severe exacerbations or ≥1 exacerbation (leading to hospital admission)

COPD Treatment & Management

Initial pharmacological treatment of stable COPD [1]

GOLD group	Daily regimen	Rescue medication
GOLD group A	• Long-acting bronchodilator (preferred if available and affordable) ◦ Long-acting beta agonist (LABA), e.g., **salmeterol**, formoterol ◦ OR long-acting muscarinic antagonist (LAMA), e.g., tiotropium bromide • OR consider a rescue medication alone for patients with very occasional dyspnea.	• Short-acting bronchodilator for immediate relief of dyspnea ◦ Short-acting beta agonist (SABA), e.g., **albuterol**, levalbuterol ◦ Or short-acting muscarinic antagonist (SAMA), e.g., ipratropium bromide
GOLD group B	• LABA/LAMA combination therapy, e.g., umeclidinium/vilanterol	
GOLD group E	• LABA/LAMA combination therapy • OR, in patients with an eosinophil count ≥ 300 cells/mcL, consider LAMA/LABA/ICS triple therapy, e.g., fluticasone furoate/umeclidinium/vilanterol	



Nonpharmacological treatment:

- **Oxygen supplementation* (improves mortality/survival outcomes)**
- Lifestyle modifications (i.e. smoking cessation)

COPD Complications

- Chronic respiratory failure
- Acute exacerbation of COPD
- Cor pulmonale (right heart failure)
- Secondary spontaneous pneumothorax due to rupture of bullae

Blue Bloater

Chronic Bronchitis



Symptoms

- Chronic, productive cough
- Purulent sputum
- Hemoptysis
- Mild dyspnea initially
- Cyanosis (due to hypoxemia)
- Peripheral edema (due to cor pulmonale)
- Crackles, wheezes
- Prolonged expiration
- Obese

Complications

- Secondary polycythemia vera due to hypoxemia
- Pulmonary hypertension due to reactive vasoconstriction from hypoxemia
- Cor pulmonale from chronic pulmonary hypertension

Pink Puffer

Emphysema



Symptoms

- Dyspnea
- Minimal cough
- Increased minute ventilation
- Pink skin, Pursed-lip breathing
- Accessory muscle use
- Cachexia
- Hyperinflation, barrel chest
- Decreased breath sounds
- Tachypnea

Complications

- Pneumothorax due to bullae
- Weight loss due to work of breathing

COPD
Chronic obstructive pulmonary disease

Epidemiology
Sex: 3:2 male/female ratio
Prevalence (US): ~6%
Third most common cause of death worldwide

Etiology/risk factors
Exogenous: tobacco smoking, air pollution, occupational exposures
Endogenous: α1-Antitrypsin deficiency, antibody deficiency syndromes, premature birth

Diagnosis

Pulmonary function tests:
Blood gas analysis
Imaging: chest x-ray, chest CT
α1-Antitrypsin if suspecting AATD

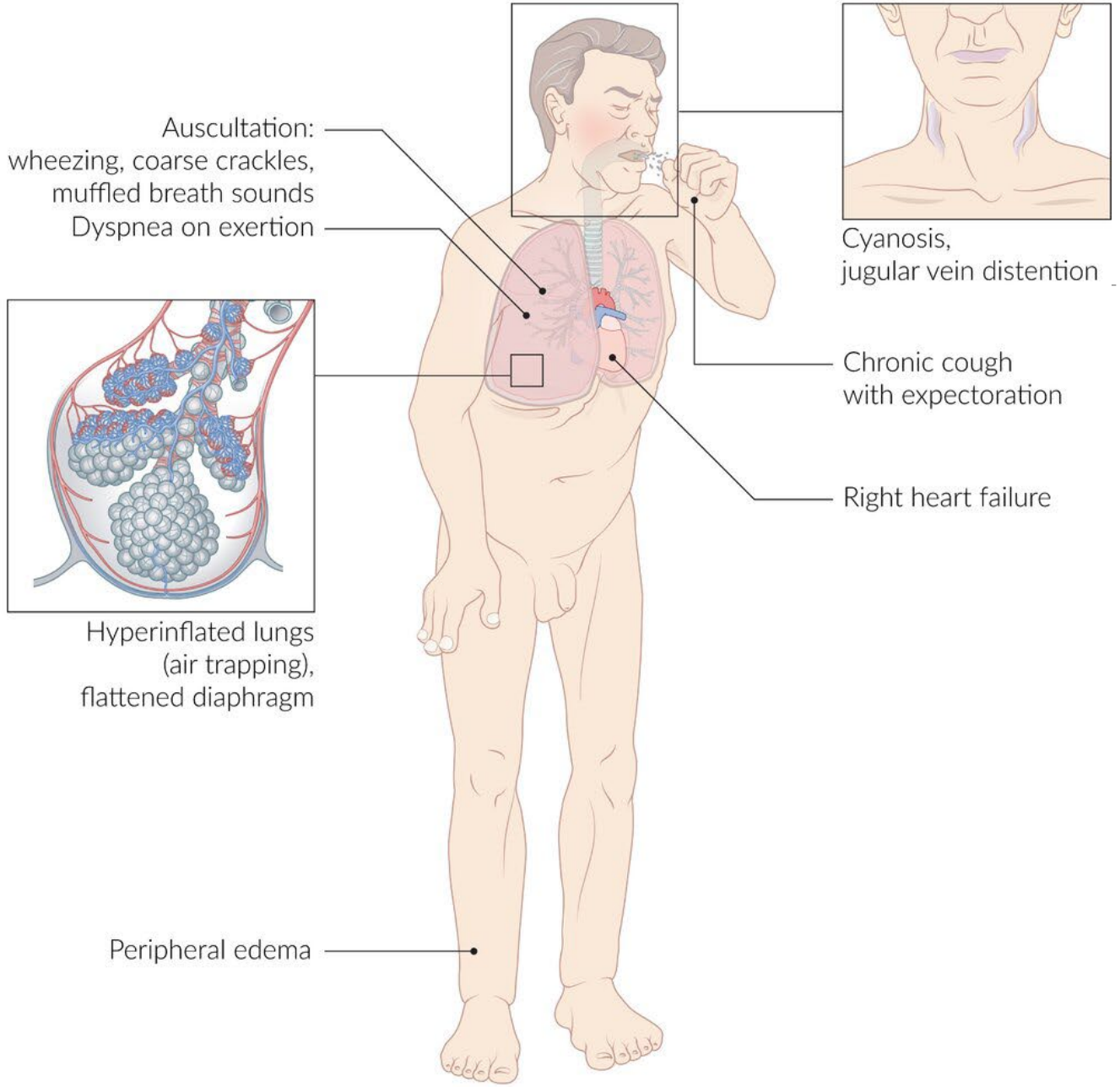
Complications
Acute exacerbations (AECOPD)
Chronic respiratory failure

Note
No curative therapy

GOLD classification

Classification based on airflow limitation in patients with $FEV_1/FVC < 70\%$

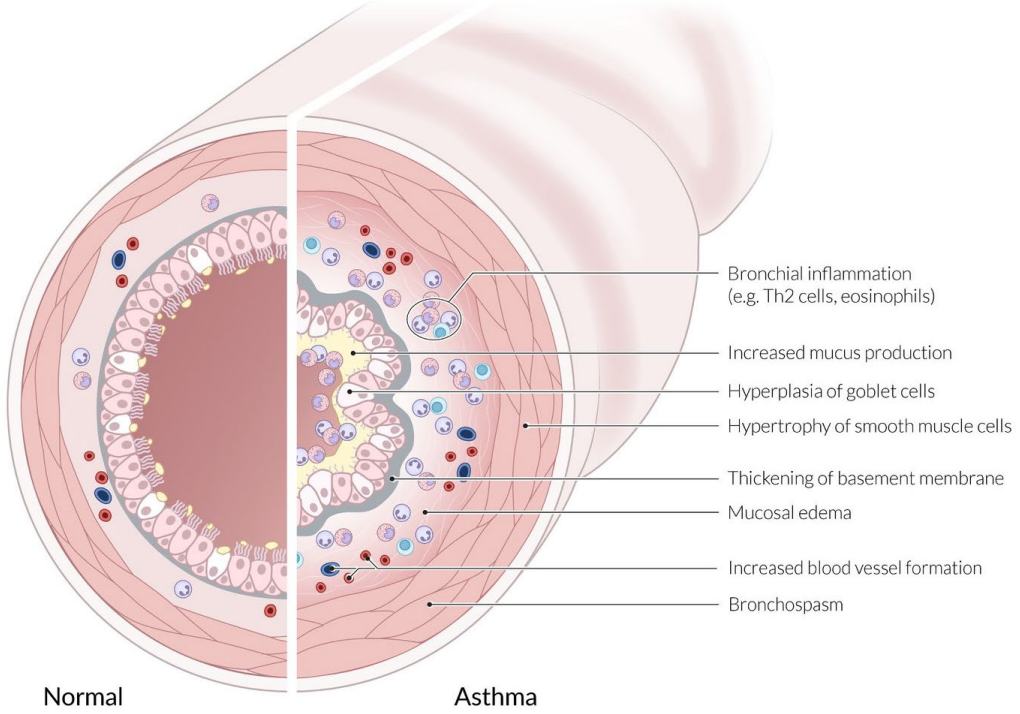
Category	Symptoms	FEV_1 % of the predicted value
GOLD 1 (Class I)	Mild	≥ 80%
GOLD 2 (Class II)	Moderate	50% ≤ FEV_1 < 80%
GOLD 3 (Class III)	Severe	30% ≤ FEV_1 < 50%
GOLD 4 (Class IV)	Very severe	< 30%



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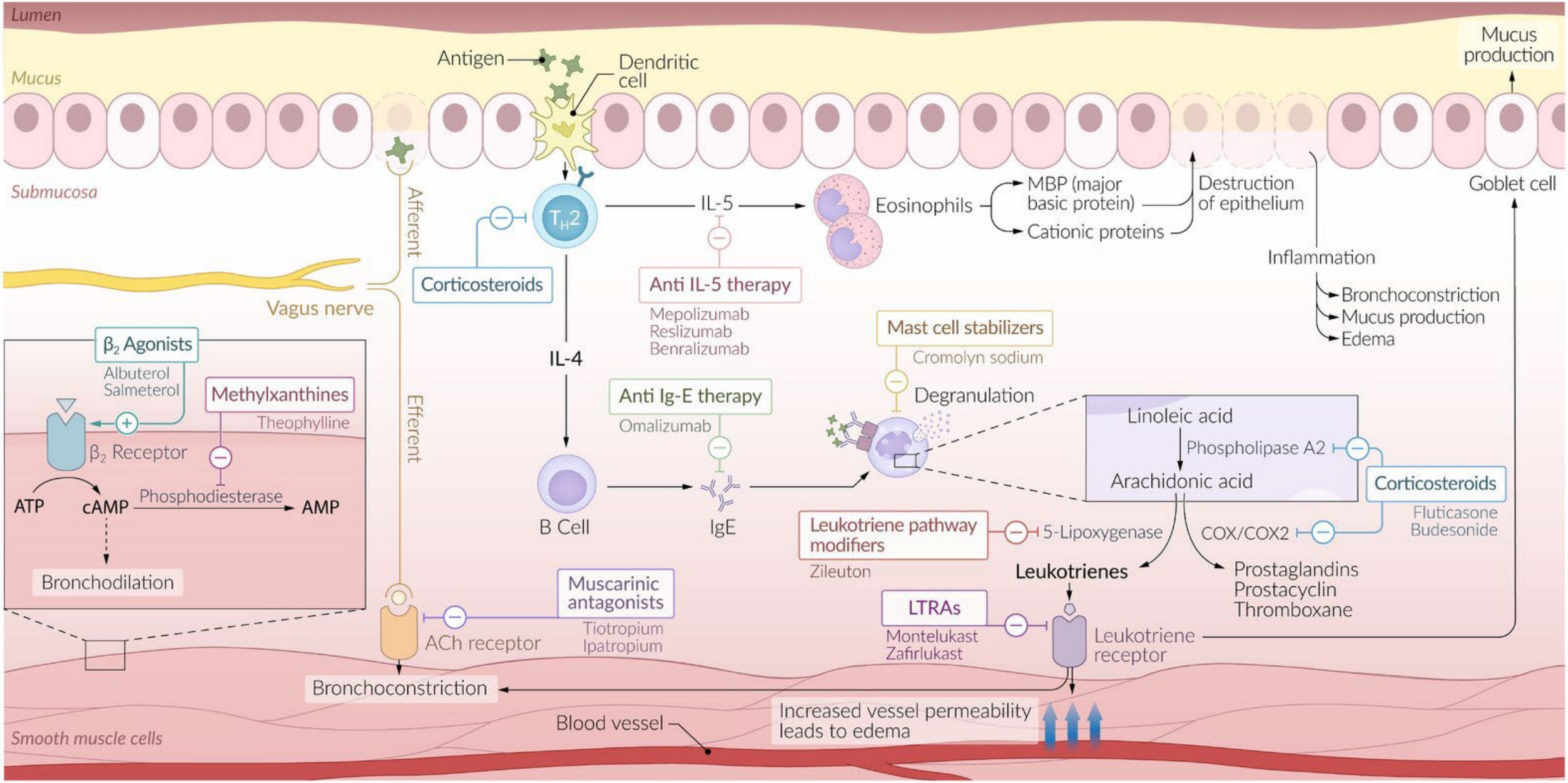
Asthma

- Unknown etiology with chronic airway inflammation driven by T-helper type 2 cells (Th2-cell)
- Pathophysiology: **Bronchial hyperresponsiveness, Bronchial inflammation, Endobronchial obstruction**
- Asymptomatic at baseline with episodes of persistent, dry cough, end-expiratory wheezing, tachypnea, dyspnea, or hypoxemia
- Allergic asthma
  - IgE-mediated type 1 hypersensitivity
  - Childhood
- Nonallergic asthma
  - Irritant or Aspirin-exacerbated respiratory disease (AERD)
  - > 40 years
- Risk factors: Atopy, Allergies, Genetic predisposition, Environmental and socioeconomic factors
- Triggers: viral URI, allergens, stress, aspirin
- Diagnosis
  - Prescence of typical asthma symptoms that vary in severity
  - Spirometry +/- bronchodilator responsiveness testing or bronchial challenge tests
  - Sputum analysis: Curschmann spirals and Charcot-Leyden crystals
- Complications: pulsus paradoxus, status asthmaticus



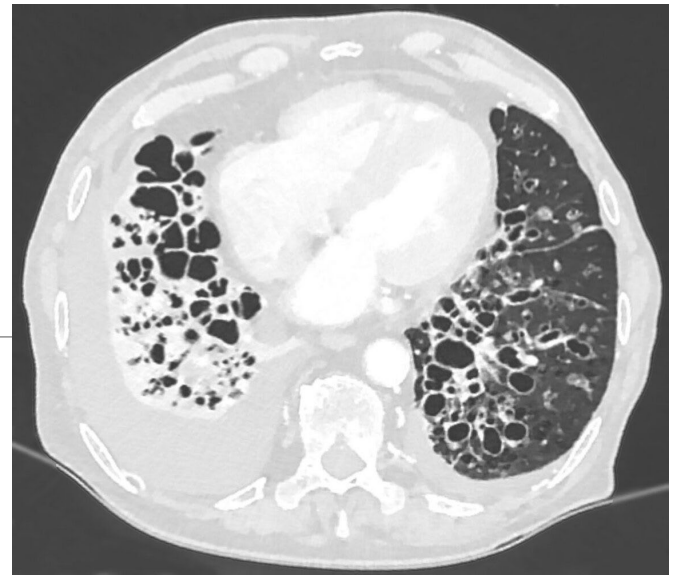
Overview of commonly used asthma medications [3][6][20]

Class	Examples	Indications and uses	Mechanism
ICS/LABA (combination of inhaled corticosteroid and long-acting beta agonist)	• Budesonide/formoterol	• Maintenance and reliever therapy	• Combination of action of ICS plus bronchodilation
	• Fluticasone/salmeterol	• Maintenance therapy	
Inhaled corticosteroids (ICS) [3][6]	• Budesonide • Fluticasone • Mometasone • Others: beclomethasone, ciclesonide, triamcinolone	• Maintenance therapy	• Inhibit transcription factors (e.g., NF-κB) → ↓ expression of proinflammatory genes (see "Glucocorticoids")
Short-acting beta-2 agonists (SABA)	• Inhaled: albuterol • Oral: terbutaline	• Reliever therapy	• Dilation of bronchial smooth muscles (see "long-acting beta agonists")
Long-acting beta-2 agonists (LABA)	• Salmeterol • Formoterol	• Maintenance therapy • Use in combination with ICS.	
Short-acting muscarinic antagonists (SAMA)	• Ipratropium bromide	• Reliever therapy for severe and life-threatening asthma exacerbations	• Competitively inhibit postganglionic muscarinic receptors in bronchial smooth muscle → bronchodilator (see "Muscarinic antagonists")
Long-acting muscarinic antagonists (LAMA)	• Tiotropium bromide	• Maintenance therapy • Use in combination with ICS.	
Oral glucocorticoids	• Methylprednisolone • Prednisone burst [3] • Equivalent dose of other oral glucocorticoids	• Severe persistent asthma • Management of acute asthma exacerbations • 3–10 day burst to achieve control during a period of deterioration	• Inhibit transcription factors (e.g., NF-κB) → ↓ expression of proinflammatory genes (see "Glucocorticoids")
Leukotriene receptor antagonists (LTRAs)	• Montelukast • Zafirlukast	• Exercise-induced asthma • Aspirin-induced asthma • Maintenance therapy (particularly in children)	• Prevent leukotrienes from binding to their G protein-coupled receptors (CysLT1) → ↓ bronchoconstriction and inflammation


Lumen
Mucus
Submucosa
Antigen
Dendritic cell
Mucus production
Goblet cell
Afferent
Efferent
Vagus nerve
T_H2
IL-5
Eosinophils
MBP (major basic protein)
Cationic proteins
Destruction of epithelium
Inflammation
Bronchoconstriction
Mucus production
Edema
Corticosteroids
Anti IL-5 therapy
Mepolizumab
Reslizumab
Benralizumab
IL-4
Mast cell stabilizers
Cromolyn sodium
Degranulation
Anti Ig-E therapy
Omalizumab
B Cell
IgE
β_2 Agonists
Albuterol
Salmeterol
Methylxanthines
Theophylline
β_2 Receptor
ATP
cAMP
Phosphodiesterase
AMP
Bronchodilation
Linoleic acid
Phospholipase A2
Arachidonic acid
Corticosteroids
Fluticasone
Budesonide
Leukotriene pathway modifiers
Zileuton
5-Lipoxygenase
COX/COX2
Leukotrienes
Prostaglandins
Prostacyclin
Thromboxane
LTRAs
Montelukast
Zafirlukast
Leukotriene receptor
Muscarinic antagonists
Tiotropium
Ipatropium
ACh receptor
Bronchoconstriction
Blood vessel
Increased vessel permeability leads to edema
Smooth muscle cells

Bronchiectasis



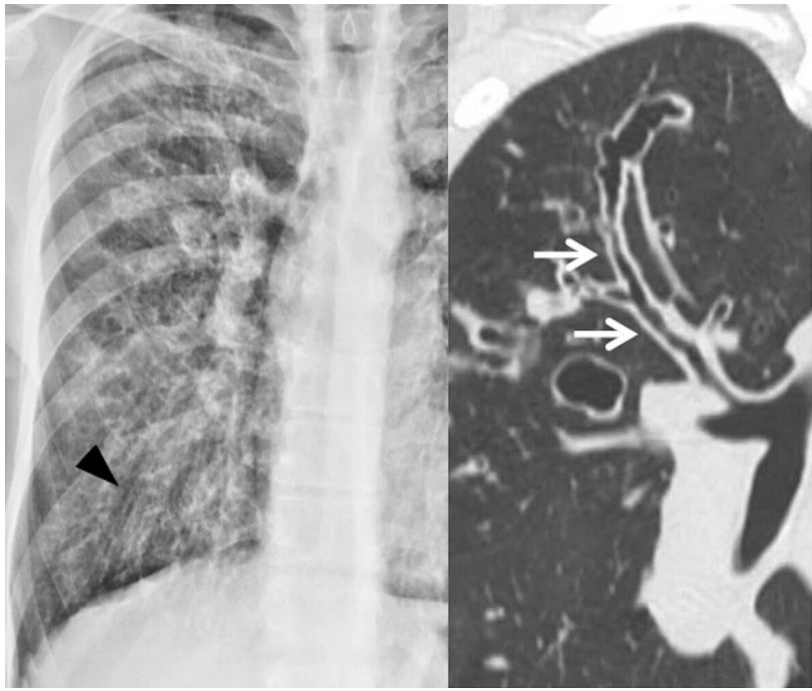


•Chronic necrotizing infection of bronchi or obstruction leads **to permanently dilated large airways** and thickened bronchial walls of small/medium airways

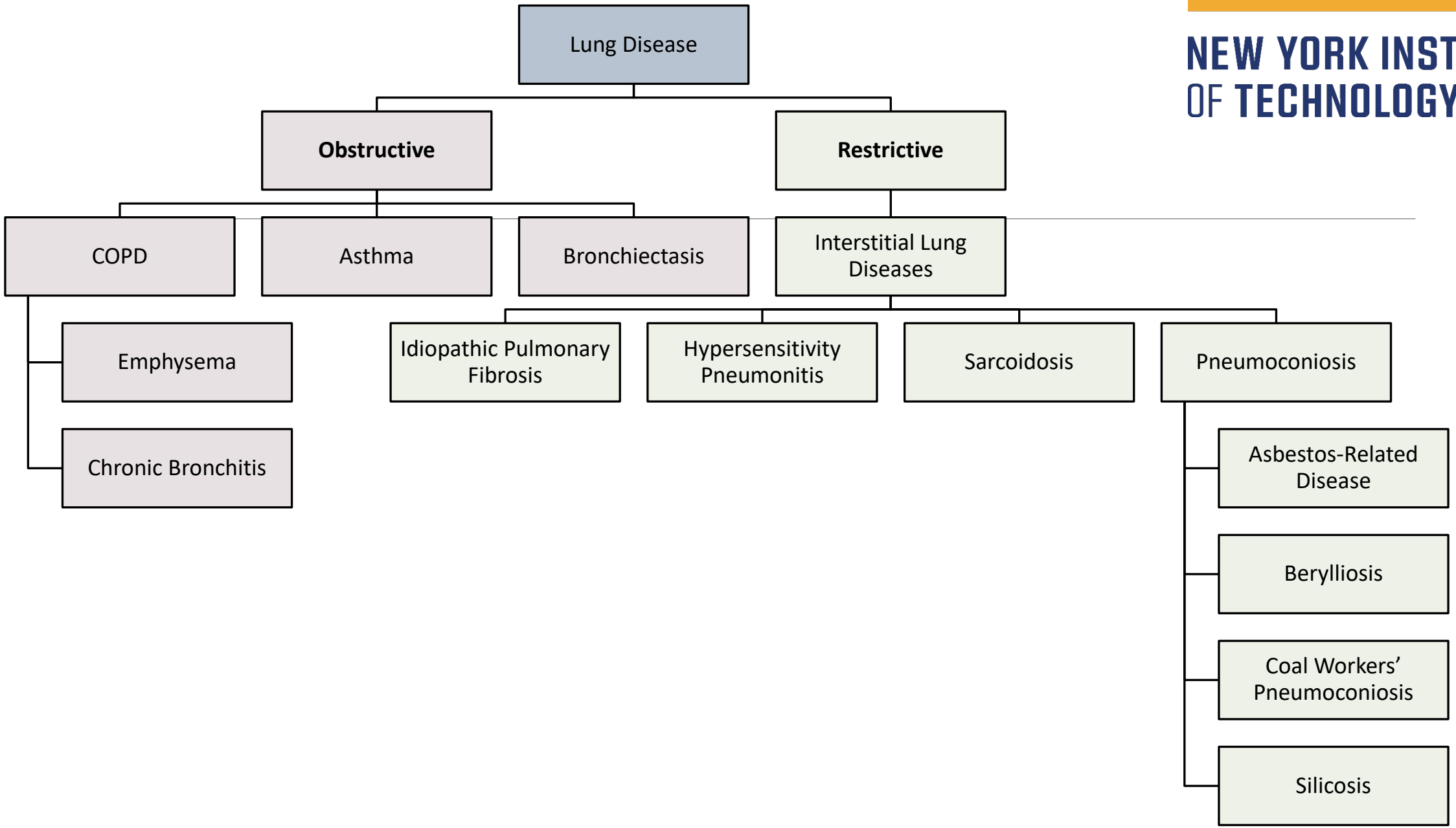
•Presents **with chronic (months-years), daily, copious, purulent sputum**, hemoptysis, digital clubbing, recurrent infections (*P. aeruginosa*)

•Associated with bronchial obstruction, poor ciliary motility (tobacco smoking, Kartagener syndrome), cystic fibrosis, allergic bronchopulmonary aspergillosis (ABPA), pulmonary infections (*M. avium*)

- Diagnosis: CXR (initial) and HRCT (confirmatory)
  - Bronchial dilation
  - Cylindrical or tubular (most common): parallel tram track sign and signet ring sign
  - Varicose
  - Saccular or cystic (most severe form)
  - Thickened bronchial walls, mucus plugging, honeycombing (suggests late-stage)
- Treatment
  - Acute
    - Supportive treatment and oxygen therapy,
    - Mucoactive agents and airway clearance techniques
    - Obtain a new sputum culture and start empiric antibiotic therapy (tailor to the most recent sputum culture)
  - Long term: stop/delay disease progression, reduce exacerbation frequency (goal ≤ 2 per year), achieve symptom control, and improve patient's QOL
- Complications:
  - Recurrent bronchopulmonary infections → chronic obstructive pulmonary disease → respiratory failure and cor pulmonale
  - Pulmonary hemorrhage (massive hemoptysis)
  - Lung abscess



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Patient presents with long term shortness of breath and undergoes spirometry testing, which reveals decreased total lung capacity and a normal forced expiratory volume in one second (FEV1) to forced vital capacity (FVC) ratio.

Is this patient presenting with obstructive or restrictive lung disease pattern?

What is on your differential for this classification of lung disease?

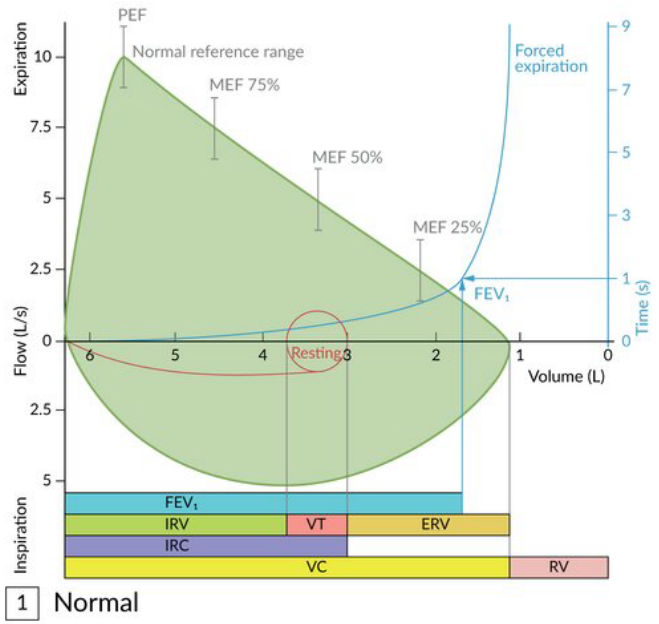
Patient presents with long term shortness of breath and undergoes spirometry testing, which reveals decreased total lung capacity and a normal forced expiratory volume in one second (FEV1) to forced vital capacity (FVC) ratio.

Is this patient presenting with obstructive or restrictive lung disease pattern?
Restrictive

What is on your differential for this classification of lung disease?
Interstitial Lung Diseases including Idiopathic Pulmonary Fibrosis, Hypersensitivity Pneumonitis, Sarcoidosis, Pneumoconiosis

Flow-Volume Loops

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1 Normal

PEF = Peak expiratory flow

MEF = Mean expiratory flow 75 / 50 / 25%

FEV_1 = Forced expiratory volume in 1 second

IRV = Inspiratory reserve volume

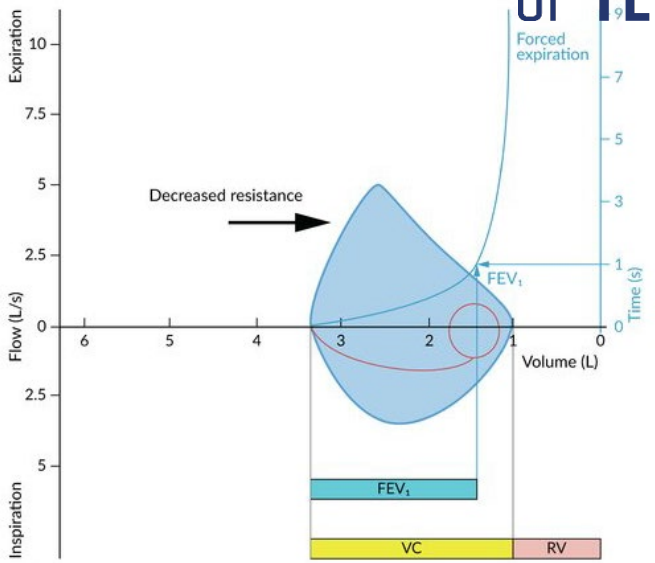
VT = Tidal volume

ERV = Expiratory reserve volume

IRC = Inspiratory reserve capacity

VC = Vital capacity

RV = Residual volume



3 Restriction

Flow-Volume Parameter	Obstructive Lung Disease	Restrictive Lung Disease
RV	↑	↓
FRC	↑	↓
TLC	↑	↓
FEV1	↓↓	↓
FVC	↓	↓
FEV1/FVC	↓ [FEV1 decreased more than FVC]	Normal or ↑ [FEV1 decreased proportionately to FVC]

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Idiopathic Pulmonary Fibrosis

- **Most common type of ILD**
- Irreversible pulmonary fibrosis and impaired pulmonary function with unknown etiology
- Associated with cigarette smoking, environmental pollutants, genetic defects
- Affects most commonly men 50–70 years of age
- Presents with progressive dyspnea, fatigue, nonproductive cough, crackles, clubbing
- Diagnosis of exclusion:
  - HRCT: **Peripheral reticular, ground-glass opacities with traction bronchiectasis +/- honeycomb appearance**
  - **Interstitial pneumonia (UIP)** pattern on histology
- Complications: pulmonary hypertension, RHF, arrhythmias, CAD, respiratory failure (usually within 3-7 years), lung cancer



Hypersensitivity Pneumonitis (extrinsic allergic alveolitis)

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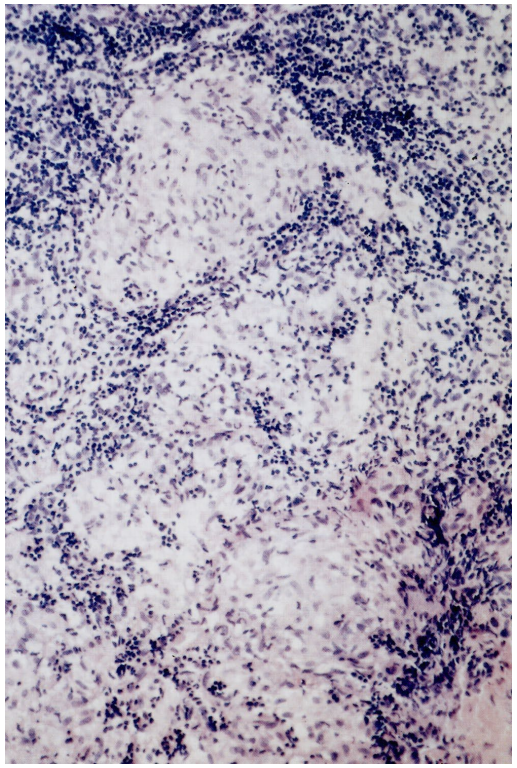
- Inhalation of **environmental and/or occupational antigens** (Farmer's lung, Pigeon breeder's lung, Chemical worker's lung)
- Combined **type III and type IV hypersensitivity reaction** with genetic predisposition
- Presents acutely with dyspnea, cough, chest tightness, fever, headache
- Diagnosis:
  - HRCT: **Honeycombing** (irreversible fibrotic changes), **thickening of alveolar septa**, **traction bronchiectasis or bronchiolectasis**, **ground-glass opacities** with inspiratory mosaic attenuation
  - Bronchoalveolar lavage (BAL): lymphocytosis (e.g., ≥ 20–30%)
  - Biopsy: **noncaseating granulomas** with lymphocytes and polynuclear giant cells
- Complications: Respiratory failure & Pulmonary heart disease
- Treatment:
  - Antigen avoidance & supportive management
  - Pharmacotherapy
    - Immunosuppressants (Glucocorticoid therapy, mycophenolate mofetil, azathioprine, rituximab)
    - Antifibrotic drugs (e.g., nintedanib, pirfenidone)



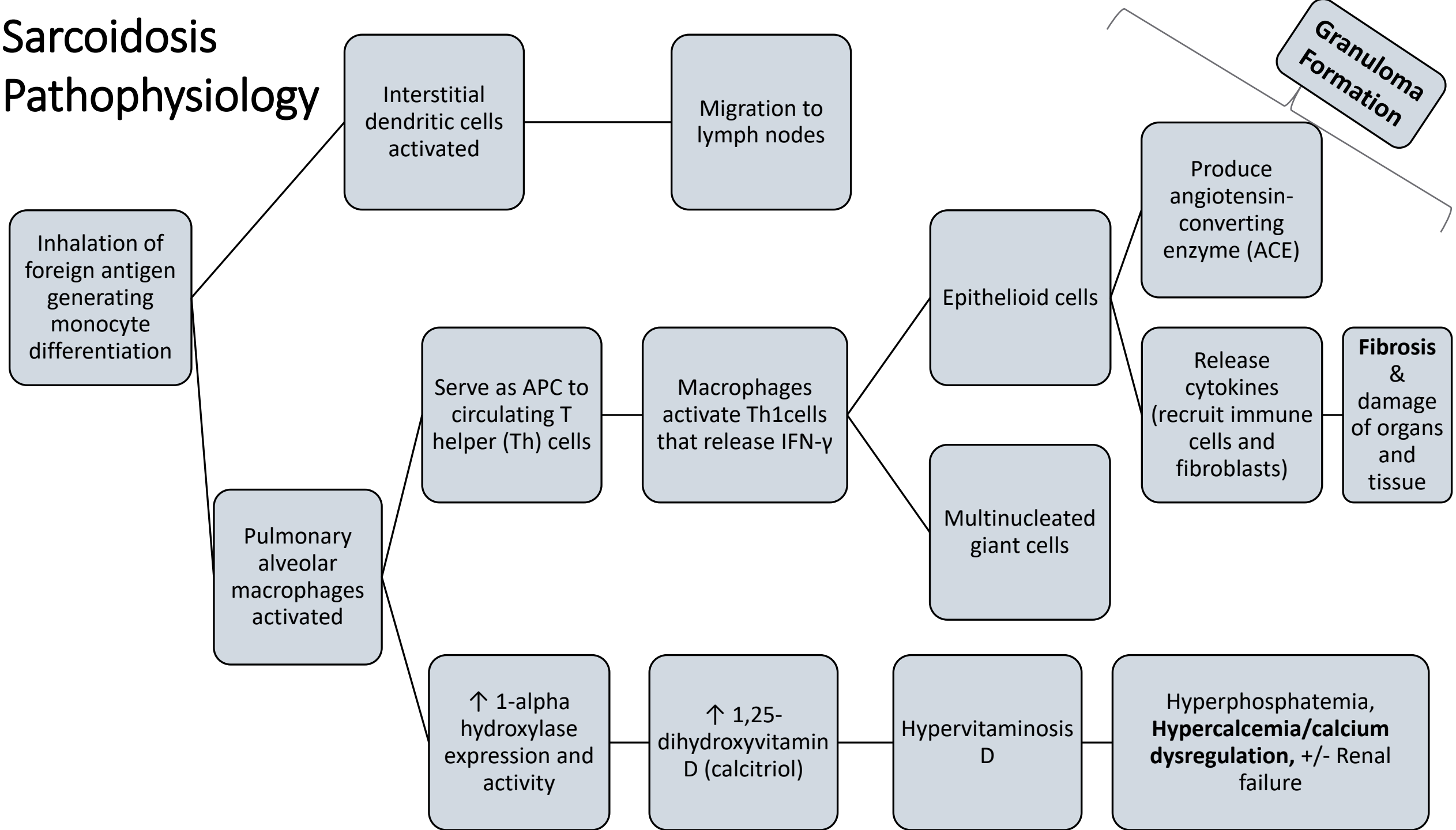
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Sarcoidosis

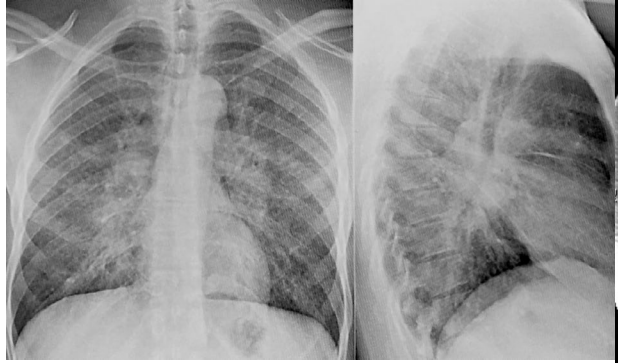
- Unknown etiology but is thought to be related to an environmental exposure in a genetically predisposed individual
- Affects most commonly **African American or Scandinavian females 30–55 years of age**
- Symptomology:
  - Variable presentation from asymptomatic (early) to progressive, relapsing disease.
  - **Symptoms can include fever, malaise, lack of appetite, weight loss, lymphadenopathy, dyspnea, cough, chest pain, arthritis, erythema nodosum, anterior uveitis**
- Systemic disorder characterized by widespread, immune-mediated formation of **noncaseating granulomas**
  - Differentiation and clonal expansion of T cells/T cell recruitment and ↑ B-cell activity → local immune hyperactivity and inflammation → formation of noncaseating granulomas in the lungs and the lymphatic system

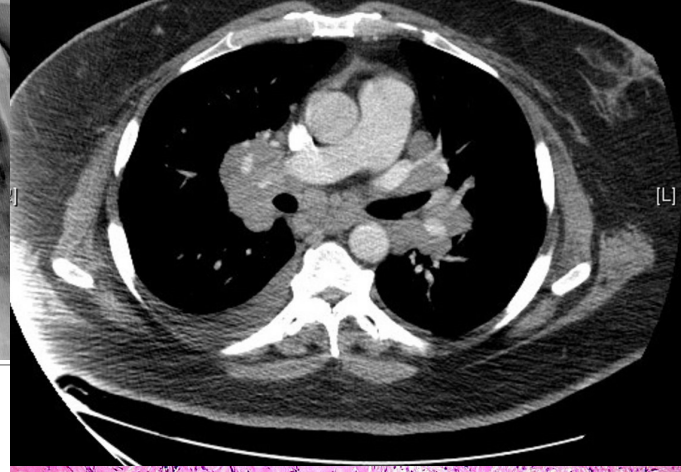


Sarcoidosis Pathophysiology



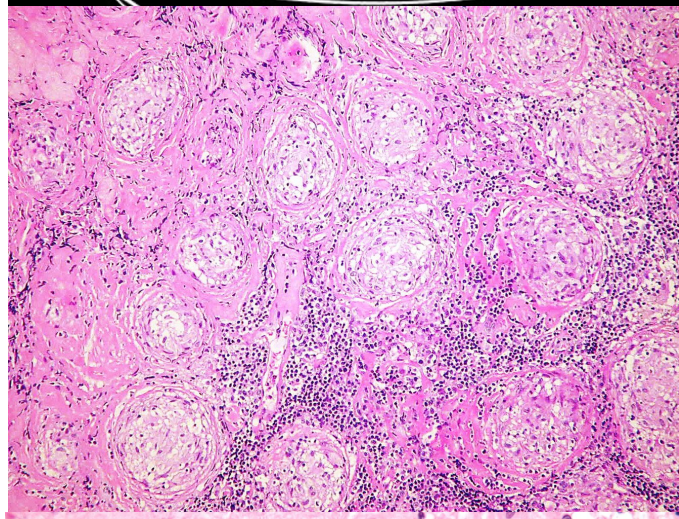
Sarcoidosis

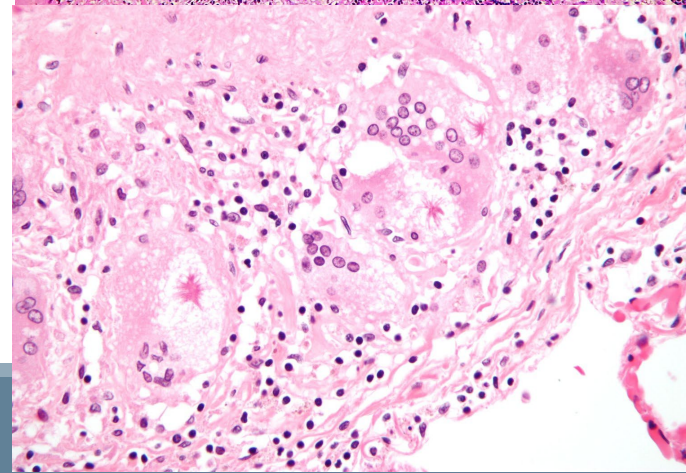




Diagnosis of exclusion:

- **Noncaseating granulomas** on biopsy
- Labs: leukocytopenia, ↑ESR, ↑CRP, **↑serum ACE levels, hypercalcemia, hypercalciuria,** ↑ IgG
- CXR: **bilateral hilar lymphadenopathy** with or without pulmonary infiltrates (reticular and/or ground glass opacities)
- HRCT:
  - **Extensive bilateral hilar and mediastinal lymphadenopathy**
  - Bronchocentric and perilymphatic micronodules (i.e., bronchovascular beading)
  - Peribronchial thickening
  - Fibrosis
- BAL:**↑ CD4:CD8 ratio**
- Biopsy [Gold standard]: Histopathology findings of **noncaseating granulomas with giant cells, Asteroid bodies, Schaumann bodies**





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Sarcoidosis

- Treatment
  - Supportive therapy: NSAIDs for pain relief, noninvasive ventilation, supplemental oxygen
  - Pharmacological therapy (immunosuppressive therapy) indicated for patients with symptomatic progressive disease, persistent pulmonary infiltration, or progressive decline of lung function
    - First line: **Glucocorticoids** (prednisone)
    - +/- DMARDs
      - Methotrexate or azathioprine
      - Chloroquine or hydroxychloroquine
      - Biologics (TNF-alpha inhibitors)
- Complications:
  - Malignancy (lung cancer, malignant lymphomas)
  - Bronchiectasis
  - Pulmonary hypertension
  - Aspergilloma
  - Chronic renal failure



Sarcoidosis

Etiology
Multisystem disorder with T-cell dysfunction and noncaseating granuloma formation

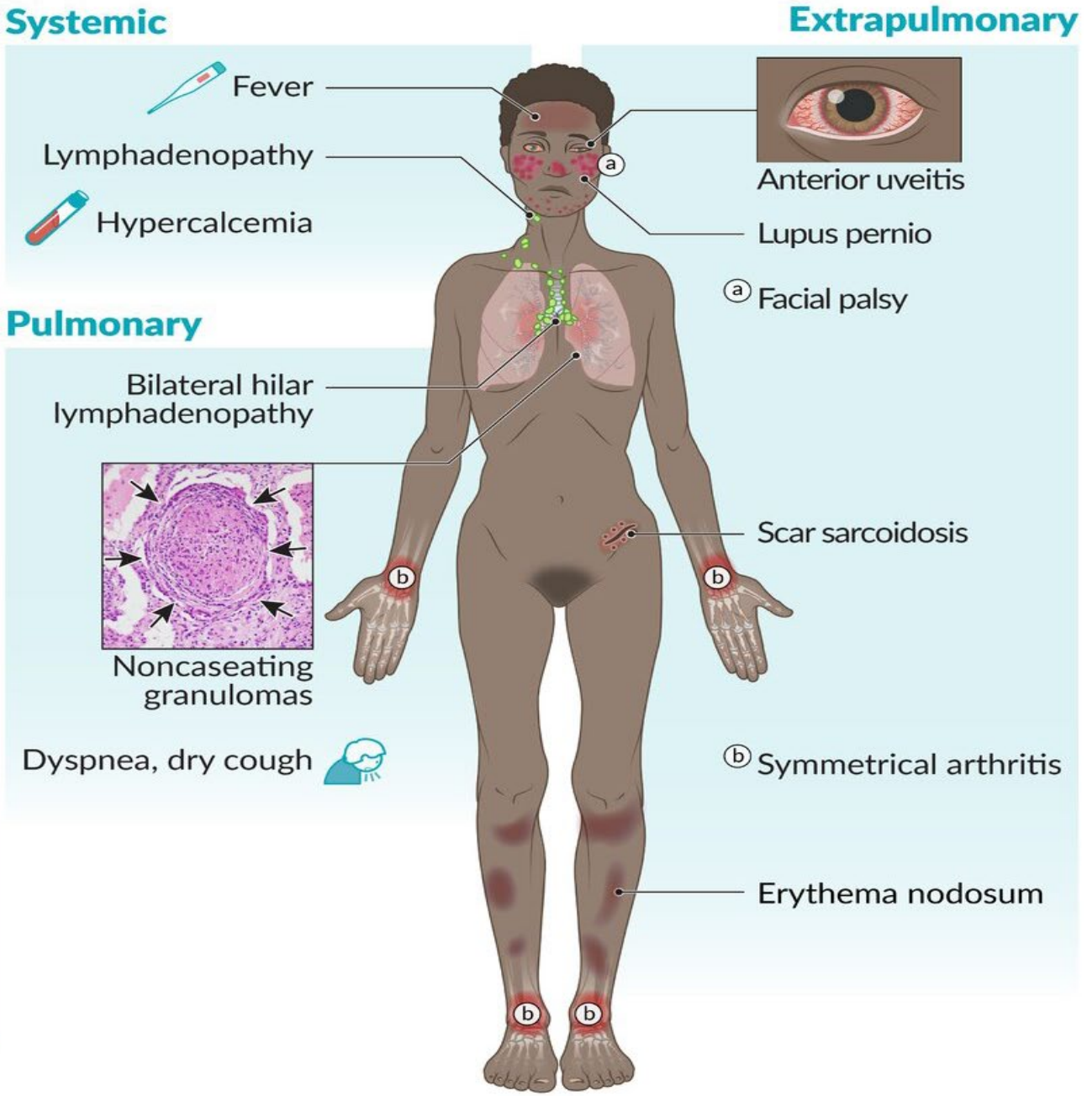
Epidemiology
♀ > ♂; peak age: 30–55 years
Incidence highest in the US among African American women

Histopathology (gold standard)
Noncaseating granulomas on biopsy; exclude more common granulomatous diseases

Diagnostic studies
- Chest x-ray: bilateral hilar lymphadenopathy
- Bronchoalveolar lavage: CD4:CD8 ratio > 3.5 supports the diagnosis

Complications
Bronchiectasis, pulmonary fibrosis, chronic renal failure

Prognosis
Spontaneous remission in 20%–70% of patients



Pneumoconiosis: Asbestos-Related Disease

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Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Asbestos-Related Disease	Airborne asbestos fibers	Asbestos miners and millers Brake linings and insulation manufactures Ship construction workers Demolishers Firefighters Plumbers Roofers	Symptoms typically develop 20–30 years after initial exposure Presents with exertional dyspnea, cough, digital clubbing Microscopic **asbestos bodies (ferruginous body)** in alveolar septa on histology Complications: Lung cancer (bronchogenic carcinoma > mesothelioma), Caplan syndrome	Diffuse bilateral infiltrates (**lower lobes**) Interstitial fibrosis **"Ivory white" calcified subdiaphragmatic and pleural plaques** (pathognomonic) Pleural effusion

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Pneumoconiosis: Silicosis

Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Silicosis	Crystalline silica dust Macrophages release fibrogenic factors in response to silica	Miners Workers of quarries, pottery production, sandblasting, glass manufacturers Construction workers	Acute silicosis: Dyspnea, cough, pleuritic pain Chronic silicosis: Chronic cough, Exertional dyspnea, Fatigue, Weight loss Complications: **Tuberculosis**, Progressive massive fibrosis, Respiratory failure, Cor pulmonale, Caplan syndrome	**Eggshell calcifications** of hilar lymph nodes Rounded, solitary, small opacities (**upper lobes**) Bilateral diffuse ground-glass opacities

Pneumoconiosis: Coal Workers' Pneumoconiosis

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Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Coal Workers' Pneumoconiosis (Black Lung Disease)	Coal dust **Carbon-laden macrophages**	Urban population Coal miners	Chronic bronchitis that progresses to progressive massive pulmonary fibrosis Complications: Caplan syndrome, Anthracosis	Fine nodular opacifications (**upper lungs**)

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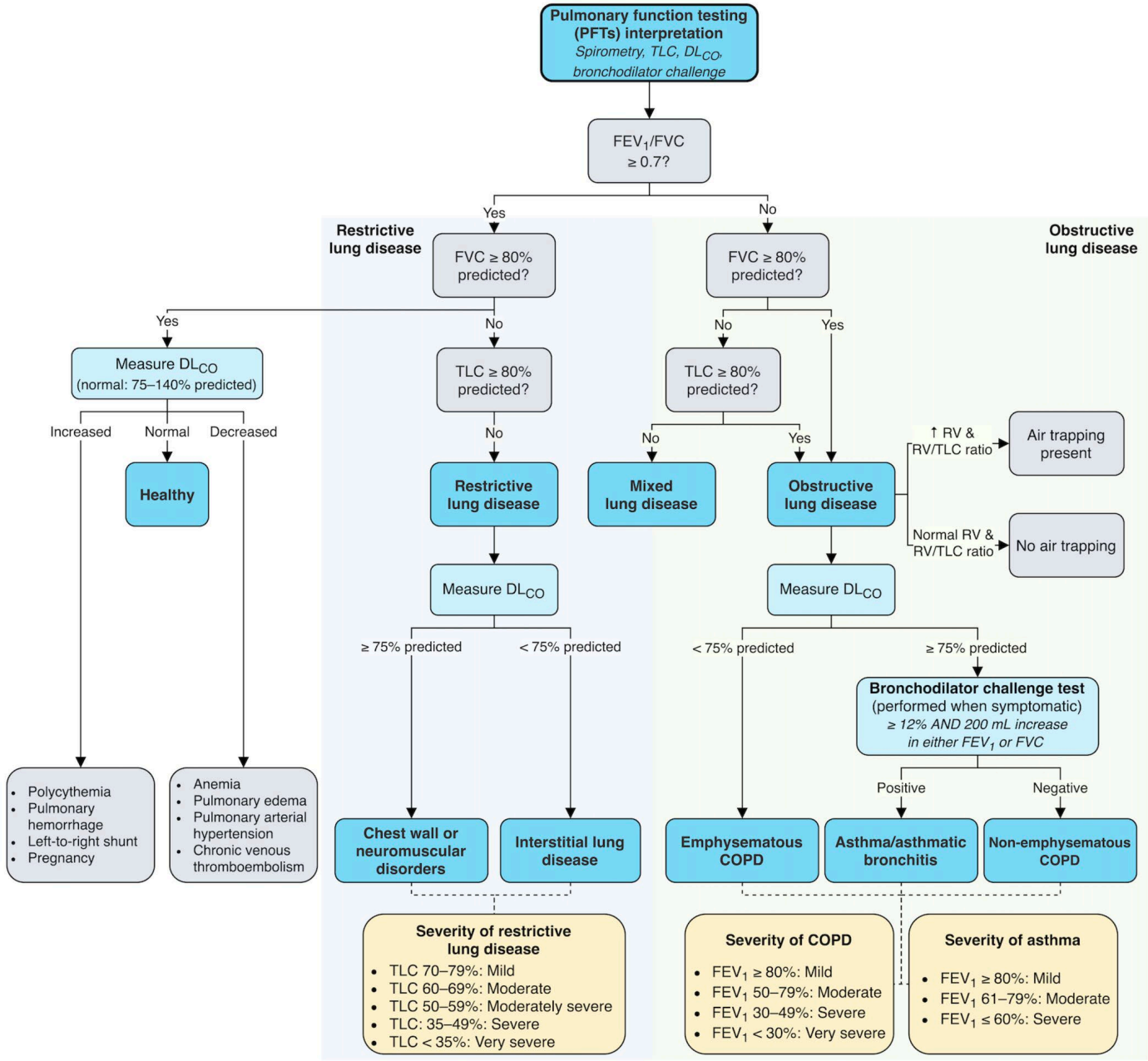
Pneumoconiosis: Berylliosis

Pathophysiology: inhalation of dust particles → phagocytosis by alveolar macrophages → destruction of alveolar macrophages, inflammatory reaction → scarring, granuloma formation

	Etiology	Population at Risk	Clinical Presentation	Chest X-Ray Findings
Berylliosis	Beryllium	Aerospace engineering Ceramics industries Dental material production Nuclear and electronics plants Dye manufacturing Foundries	Progressive dyspnea within a few days of high-grade exposure **Noncaseating granulomatous** disease Complications: lung cancer, Cor pulmonale	Reticular interstitial pattern (**upper lobes**) +/- hilar lymphadenopathy

Summary

- Obstructive and restrictive lung diseases have different finding on spirometry and flow-volume loops that aid in diagnosis
- Symptomology and clinical presentations may overlap for obstructive and restrictive lung diseases; thus, it is important to know the diagnostic testing as well as findings
- The etiology and pathophysiology of obstructive and restrictive lung diseases can help explain symptomology
- Treatment modalities include pharmacological and non-pharmacological management and are important to be aware of
- Various, serious complications can arise from obstructive and restrictive lung diseases



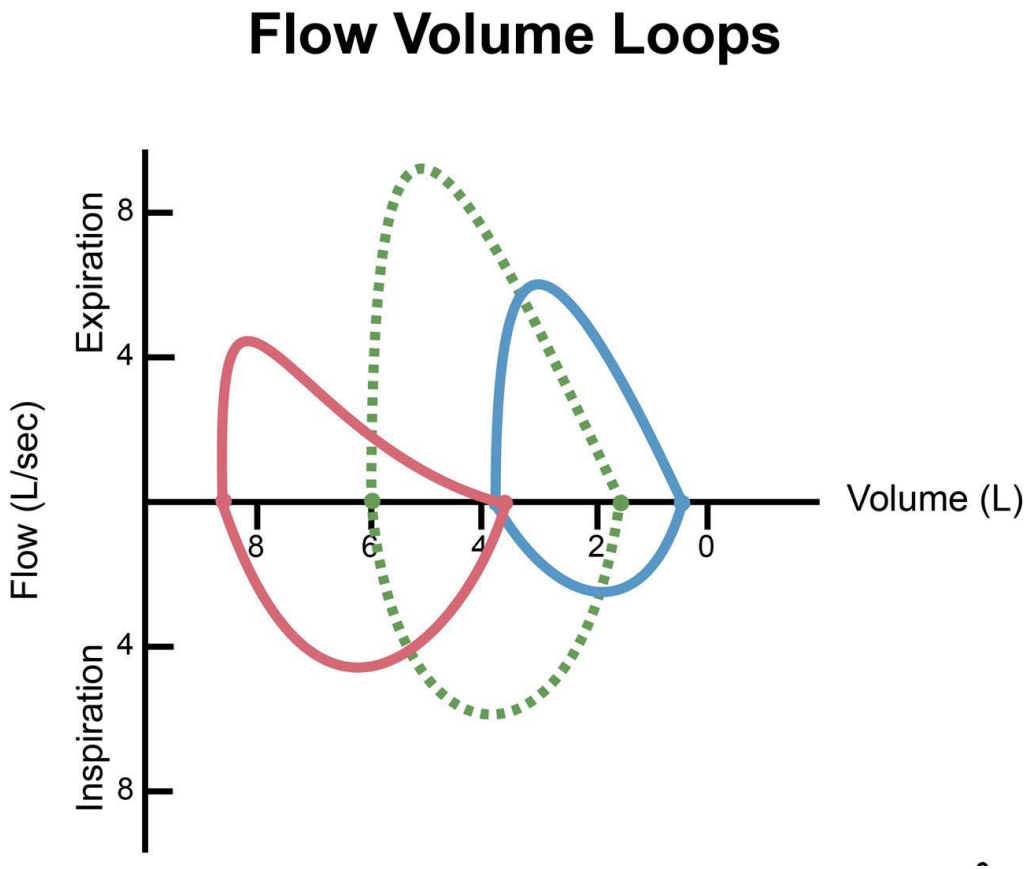
Practice Questions

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Practice Question

A 56-year-old patient presents with dyspnea that has been occurring for 2 years getting progressively worse. On social history he endorses smoking 3 packs per day for 30 years. Chest x-ray reveals an increased anterior-posterior diameter with a flattened diaphragm. After stabilization with oxygen and pharmacological therapy he is referred to follow up at an outpatient pulmonologist. At his follow up, he undergoes spirometry testing.

Which color flow-volume loop would you expect to see in this patient?



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Practice Question

A 4-year-old boy is brought to the physician in December for episodic shortness of breath and a nonproductive cough for 3 months. These episodes frequently occur before sleeping, and he occasionally wakes up because of difficulty breathing. His mother also reports that he became short of breath while playing with his friends at daycare on several occasions. He is allergic to peanuts. He is at the 55th percentile for height and weight. Vital signs are within normal limits. Examination shows mild scattered wheezing in the thorax. An x-ray of the chest shows no abnormalities.

1. Which of the following is the most likely diagnosis?
   a) Cystic fibrosis
   b) Foreign body aspiration
   c) Asthma
   d) Chronic bronchitis
   e) Interstitial lung disease

2. What findings would most likely be seen on spirometry in this patient?
   a) Decreased FEV1
   b) Increased FEV1/FVC ratio
   c) Decreased TLC
   d) Decreased RV
   e) Normal FRC

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Practice Question

A 65-year-old woman with chronic obstructive pulmonary disease comes to the emergency department with a 2-day history of worsening shortness of breath and cough. She has a mild productive cough for the past 2 years but she noticed that her sputum is more yellow than usual over the past 2 days. She has not had any recent fevers, chills, sore throat, or a runny nose. Her only medication is a salmeterol inhaler that she uses twice daily. Her temperature is 36.7°C (98.0°F), pulse is 104/min, respirations are 28/min, and blood pressure is 134/73 mm Hg. Pulse oximetry shows an oxygen saturation of 85% on room air. She appears uncomfortable and shows labored breathing. Auscultation of the lungs discloses rhonchi that disappear after coughing and diffuse bilateral wheezes. A chest x-ray shows mild hyperinflation and flattening of the diaphragm but no consolidation. She is started on supplemental oxygen via nasal cannula.

Which of the following is the most appropriate initial pharmacotherapy?

a) Prednisone and roflumilast

b) Albuterol and montelukast

c) Prednisone and salmeterol

d) Albuterol and theophylline

e) Prednisone and albuterol

f) Prednisone and tiotropium

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Practice Question

A 33-year-old man is seen in the emergency department with pleuritic chest pain and shortness of breath. He has no history of serious illness and does not smoke but has worked for a pottery and glass manufacturing company for the last 10 years. An x-ray of the chest shows eggshell calcifications of the hilar lymph nodes and diffuse ground glass opacities in the upper lobes.

1. This patient is most likely to have which of the following diagnosis?
   a) Chronic bronchitis
   b) Berylliosis
   c) Silicosis
   d) Panacinar emphysema
   e) Asthma

2. This patient is at increased risk of developing which of the following long-term infectious complications?
   a) Mycobacterium avium complex
   b) Mycobacterium tuberculosis
   c) Pneumonia
   d) Aspergillosis
   e) Brucellosis

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Practice Question

A 27-year-old woman comes to the physician because of a 2-month history of increasing shortness of breath and a nonproductive cough. She has been unable to perform her daily activities. She has also had malaise and bilateral ankle pain during this period. She first noticed the symptoms after returning from a vacation to Arizona. She is a research assistant at the university's geology laboratory. She was born in Nigeria and visits her family there twice a year; her last trip was 3 months ago. Her temperature is 37.8°C (100.0°F), pulse is 100/min, respirations are 24/min, and blood pressure is 112/72 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. There is no palpable cervical or axillary lymphadenopathy. The lungs are clear to auscultation. Examination shows ciliary injection and photophobia of the left eye. The remainder of the examination shows no abnormalities. A complete blood count is within the reference range. An x-ray of the chest is shown.



1. Which of the following is the most likely diagnosis?
   a) Pulmonary tuberculosis
   b) Hypersensitivity pneumonitis
   c) Bronchiectasis
   d) Sarcoidosis
   e) Centrilobular emphysema

2. Which of the following additional findings is most likely to be found in this patient?
   a) Positive tuberculin skin test
   b) Low serum angiotensin-converting enzyme activity
   c) Elevated serum parathyroid hormone concentration
   d) Elevated CD4+ T-cell count in bronchoalveolar lavage
   e) Positive anti-dsDNA antibody testing

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Resources

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